SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes
QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I VELAN R (192512502) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS

1. Smart Home Automation System Using Intelligent Agents

A smart home automation system manages **lighting, temperature, and security** by observing user behavior and environmental conditions such as motion, temperature, and time. Different types of intelligent agents perform different functions in such a system.

1. Simple Reflex Agent

- Makes decisions based only on the current input.
- Uses condition-action rules.
- Does not remember previous states.

Example:

- If motion is detected → Turn ON the lights.
- If smoke is detected → Trigger the fire alarm.

Advantages

- Fast response
- Easy to implement

Disadvantages

- Cannot learn or adapt.
- Fails in complex situations.

2. Model-Based Reflex Agent

- Maintains an internal model of the environment.
- Uses both current and previous information.

Example:

- Remembers whether someone is already inside the house.
- Keeps lights ON while the room is occupied.
- Adjusts AC based on previous temperature changes.

Advantages

- Better decision making
- Works in partially observable environments

Disadvantages

- Requires memory and more computation.

3. Goal-Based Agent

- Selects actions that achieve a specific goal.

Example Goals

- Maintain room temperature at **24°C**
- Keep the house secure
- Minimize electricity usage

The agent evaluates multiple actions and selects the one that best achieves these goals.

Advantages

- Flexible
- Can plan ahead

Disadvantages

- Requires more processing time.

4. Utility-Based Agent

- Chooses the action that provides the highest overall benefit (utility).

The utility function considers:

- Comfort
- Energy efficiency
- Security
- Cost

Example
If electricity prices are high:

- Slightly reduce AC usage
- Use natural lighting
- Maintain acceptable comfort while reducing electricity cost.

Advantages

- Produces optimal decisions
- Balances multiple objectives

Disadvantages

- Designing the utility function is difficult.

Comparison

Agent	Memory	Planning	Learning	Example
Simple Reflex	No	No	No	Turn ON lights when motion detected
Model-Based	Yes	Limited	No	Maintain room occupancy status
Goal-Based	Yes	Yes	Limited	Keep home secure and comfortable
Utility-Based	Yes	Yes	Yes (can be combined)	Optimize comfort, security and energy consumption

Most Effective Approach

A **hybrid approach combining Model-Based and Utility-Based Agents** is the best choice.

Reason:

- Stores information about previous states.
- Learns user preferences.
- Optimizes comfort and energy efficiency.
- Maintains high security.
- Makes intelligent decisions under different situations.

Conclusion

A hybrid **Model-Based + Utility-Based Agent** provides the most effective smart home automation system because it can remember previous events, adapt to changing conditions, optimize energy usage, and improve user comfort and security.

2. Robot Navigation in an Unknown Maze Using Uninformed Search

A robot placed in an unknown maze must find the exit without prior knowledge of the environment. Two uninformed search algorithms suitable for this problem are **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS).**

1. Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the **lowest cumulative path cost**.

Working

1. Start from the initial position.
2. Store nodes in a priority queue.
3. Expand the least-cost node first.
4. Continue until the goal is reached.

Advantages

- Finds the least-cost path.
- Complete.
- Optimal when costs are positive.

Disadvantages

- High memory usage.
- Slow for large search spaces.

Complexity

- **Time:** $O(b^{(1+C^*/\varepsilon)})$
- **Space:** $O(b^{(1+C^*/\varepsilon)})$

where

- **b** = branching factor
- **C*** = optimal path cost
- **ε** = minimum step cost

2. Iterative Deepening Search (IDS)

IDS repeatedly performs Depth-Limited Search with increasing depth limits.

Working

1. Search depth = 0
2. If goal not found, increase depth.
3. Repeat until the goal is found.

Advantages

- Low memory usage.
- Complete.
- Suitable when solution depth is unknown.

Disadvantages

- Repeats node expansions.
- Not optimal if path costs differ.

Complexity

- **Time:** O(b^d)
- **Space:** O(bd)

where

- **b** = branching factor
- **d** = depth of solution

Comparison

Feature	UCS	IDS
Path Cost	Considers cost	Ignores cost
Completeness	Yes	Yes
Optimality	Yes	Only when costs are equal
Memory	High	Low
Speed	Slower	Faster for shallow solutions

Agent Design

Simple Reflex Agent

- Makes immediate decisions.
- Unsuitable for maze navigation.

Model-Based Agent

- Stores explored locations.
- Avoids revisiting the same path.

Goal-Based Agent

- Plans a path toward the exit.

Utility-Based Agent

- Chooses the safest and shortest path by minimizing cost.

Best Combination

The most effective solution is:

Goal-Based Agent + Uniform Cost Search (UCS)

Justification

- The goal-based agent continuously plans toward the maze exit.
- UCS guarantees the lowest-cost path.
- Suitable for unknown environments with varying movement costs.
- Produces an optimal solution while avoiding expensive paths.